

Generating Stock Market Data and Making Predictions Using GAN and Neural Networks

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Abstract

Solving time series forecasting has always been a problematic task for computer scientists with always uncertainty among the stakeholders regarding the behaviour of the time series data. With the help of machine learning and artificial intelligence, computer scientists have provided a solution by developing deep learning models which can solve time series forecasting problems. Stock market prediction is an example of time series forecasting and deep learning models can predict the stock market and help the shareholders in making a better decision of either buying, selling, or holding the share. In recent years, researchers have found that deep learning models have been showing great results and are outperforming traditional machine learning models. Hence, in our study, we decided to use multi-layered neural networks. We will be using the recurrent neural network (RNN), the long-short term memory neural network (LSTM), and the gated recurrent units neural network (GRU) during the course of our study. In our study, we will use these neural networks to predict the closing price of a stock market share. Additionally, for the stock market data, we will also use a generative adversarial network (GAN) to generate fake data. We will then use this fake-generated data to make predictions using the above-mentioned neural networks. We will then evaluate these models on different evaluation metrics and critically analyse the predictions made by our neural networks.

1. Introduction

The time series data has been always unpredictable with different trends and seasonal behaviour. In the stock market, investors face this challenge every day and a solution to this problem is provided by making use of Machine Learning and Deep Learning models. This appears to be a very complex problem and is a clear-cut example of a time series forecasting problem. Deep Learning models provide significant results in predicting stock market prices as they can learn patterns in sequential data and use them to make accurate predictions. In our study, we used the financial data of the Tata Steel stock which is registered in the National Stock Exchange of India.

We collected the financial data using an in-built Python library, yFinance which provides access to Yahoo Finance and we can easily download financial data of any registered stock using its unique id called ticker. Using yFinance, we downloaded historical data up to three years ago from Tata Steel which contained the required information regarding the share price during a trading day including the closing price. Since the stock market data varies with time and has different trends, we used historical data of not more than 3 years so that the market trend is somewhat similar and does not misguide our neural engine. Therefore, we are left with data over the past three years which caters to around 746 entries since the Indian stock market is closed on weekends and other national holidays. We then performed exploratory data analysis on our dataset to summarise and get a gist of the behaviour of our time series data. This helped us to understand the data better and have a glimpse of market trends and past behaviour of the share price.

We know that our dataset is somewhat small for our neural networks mainly because they require a large number of iterations as they have a lot of parameters to learn. If we have a small dataset running over a large number of iterations then there are chances for overfitting the model. We can avoid overfitting if we have a comparatively larger dataset. Therefore, we decided to generate artificial time series data which also aligns with the current market trends. This will help our neural network to perform better since we can now double the entries in our dataset by generating artificial stock market data. We used a generative adversarial network to generate artificial stock market data so that we can have a larger dataset for our neural networks. We tested our neural networks on this artificial data and compared the results with the real data just as the check the authenticity of our generated artificial data.

2. Related Works

We started our research by first collecting relevant and timely existing research work on the topic of stock market prediction just to understand the different methodologies adopted by researchers over the period. We analysed different studies, which helped us formulate our objectives and problem statement and find any potential gaps in the current work.

1. T. Kimoto et al. [1] suggest a modular neural network (MNN) methodology for forecasting stock market prices. This approach consists of multiple neural network modules, each of which specializes in predicting a specific aspect of the stock market, such as the overall trend, trading volume, or price. The output of each module is combined to obtain the final prediction. The authors also compare the results from this MNN methodology with the results obtained from the traditional ARIMA model. According to the results, the MNN approach outperforms the ARIMA model in terms of accuracy.
2. Y. Zhang and L. Wu [2] proposed a novel methodology of combining an improved bacterial colony optimization (BCO) algorithm and a backpropagation (BP) neural network. The authors used an improved version of the BCO algorithm that uses an adaptive mutation rate and a dynamic step size to enhance the optimization performance. The outcome shows that the suggested methodology outruns the other existing methods, such as support vector and the back propagation neural network, and has a better accuracy and thus demonstrates the potential of combining the BCO algorithm and BP neural network for stock market predictions.
3. K. Zhang et al. [6] propose an approach of using historical stock prices to generate synthetic stock price data using a GAN model. They evaluated the performance of their GAN model and compared the results with those obtained from traditional machine learning models such as support vector regression and LSTM model and as expected, the GAN model performed better than the traditional models and provided a better accuracy.
4. M. Hiransha et al. [11] propose the use of deep learning models for predicting stock market prices. They use LSTM and CNN and compare the performances of both these neural networks with conventional machine learning algorithms. The deep learning models outperformed the machine learning models in terms of accuracy, precision, and recall with LSTM yielding the best results.

3. Proposed Model

To solve our problem and make stock market predictions with accuracy and precision, we propose the use of three different multi-layered neural networks. We will build and train three different neural networks, namely the RNN, the LSTM, and the GRU over the same dataset, and then compare the performances of each of them. The comparison would be done based on the different errors including the mean absolute percentage error (MAPE). Our neural networks will also predict the stock price for the next trading day. Therefore, we will have an idea about the error percentage our model could have, which will help the stakeholders make better financial decisions.

We also used a generative adversarial network (GAN) to generate artificial stock market data which was identical to our original data collected from yahoo finance. By generating artificial stock market data, we solved the issue of overfitting the model which generally arises while building neural networks with a small dataset. Thus, by generating artificial data, we doubled our data size and decreased the chances of overfitting the model. We also evaluated our neural networks on this artificial data so that we can have a gist of its viability and stability. We also pre-processed the data thoroughly before inputting it into the neural networks to increase our models' efficiency. These steps included removing all null values, converting the data frame into 3-dimensional arrays with a shape that is accepted by our neural engine, and standard scaling of the data before and after the predictions.

4. Dataset and Pre-processing

During our literature survey, we found that some of the existing works did not mention the source of their data or that the source mentioned was not very credible. We wanted our data source to be authentic, open-source, and easily accessible and thus we decided to collect our historical stock price data from yahoo finance which contains financial data of companies registered in the stock markets across the globe. We decided to use the historical data of 'TATASTEEL' for up to 3 years. Each registered company has a unique name called a ticker using which the data can be downloaded.

Using this data, we generated artificial data using a generative adversarial network and later we also used this generated data for making predictions using our neural networks. We removed the null values and performed exploratory data analysis on our dataset to critically analyse the data. Before inputting the real data into our GAN model, we scaled the data using the sklearn python library and then converted the data into a 3-dimensional array. We applied these same steps before building our neural networks but this time we had three different 3-dimensional arrays, one for training the model, another for testing the model, and the third one for making future predictions. The model would use data up to two weeks ago to make predictions of the next day. Therefore, the third array contained historical prices for up to two weeks only.

5. Implementation

In our study, we decided to solve the stock market prediction problem, an example of a time series forecasting problem using different neural networks. After analyzing and pre-processing the data according to the neural networks, we moved on to implementing and training our neural networks.

The first task was to generate artificial stock market data which can further be used while training the RNN, LSTM, and GRU neural networks. For generating artificial data, we used TimeGAN, a generative model for synthesizing sequential data contingent on Generative Adversarial Networks (GANs) and autoregressive models. The TimeGAN model consists of four components; a generator network, a discriminator network, an embedding network, and a recovery network. Both the generator and the discriminator networks are trained simultaneously to improve the generator's ability to create rational time-series data that can deceive the discriminator network. At the same time, the discriminator network is trained to differentiate between the real and the artificial data. Once the TimeGAN model is trained, it will be used to generate synthetic time-series stock market data.

After successfully generating artificial data, we now built our neural networks. The RNN takes in a sequence of past values and uses them to predict the next value in the sequence. One time step of the input sequence is passed into the network, and at each iteration, the network adjusts its internal state depending on the input from the previous and current time steps. This allows the network to capture temporal dependencies in the data and make accurate predictions for future time steps. While the LSTM predicts the value for the subsequent time step from a series of historical observations as input. The network is trained on a set of historical data and learns to recognize patterns and relationships between the input sequence and the corresponding output sequence. The LSTM architecture enables the network to capture long-term dependencies and temporal dynamics in the data, making it well-suited for our task. GRUs are intended to solve the vanishing gradient issue that is commonly encountered when training traditional RNNs. GRUs are like LSTMs in that they also use gating mechanisms to selectively pass information between time steps. By selectively retaining or forgetting information from earlier time steps, GRUs learn dependencies that persist in time series data. This enables them to identify intricate links and patterns in the data, which they may then use to generate precise predictions for subsequent time steps.

We decided to have a similar architecture for the three neural networks to compare their performances fairly. Each neural network has four RNN, LSTM, and GRU layers respectively with each layer accompanied by a layer of dropout to reduce overfitting. Lastly, there is a dense layer with a single output unit, the final output layer of the neural network. The dense layer is used to produce a single output value or the predicted value. The following figure depicts the architecture of the GRU neural network.

Further, the below block diagram depicts all the steps taken during our course of study sequentially.

6. Results

We built three different multi-layered neural networks and made predictions and calculated the mean absolute error, max error, root mean square error, and the mean absolute percentage error for each of the

neural networks on real data as well as the artificial data. In the below tables, we have tabulated all the results that we have obtained.

Table 1
Results of Real Data

Model	MAE	RMSE	Max Error	MAPE (%)
RNN	2.13	2.80	8.63	1.99
LSTM	2.85	3.58	11.95	2.59
<i>GRU</i>	<i>1.64</i>	<i>2.18</i>	<i>7.25</i>	<i>1.53</i>

Table 2
Results of Artificially Generated Data

Model	MAE	RMSE	Max Error	MAPE (%)
RNN	3.70	3.91	7.75	2.88
LSTM	3.26	3.64	8.88	2.55
<i>GRU</i>	<i>1.14</i>	<i>1.72</i>	<i>7.76</i>	<i>0.89</i>

From the results obtained, we can conclude that the gated recurrent unit (GRU) neural network gave the best results for the real and artificial data, outperforming the other two neural networks. The results obtained on the real data and the artificial data are quite similar so therefore we can also conclude that the artificial data is much alike the real data.

Further, to deeply analyze the results we got, we visualized the performances of each neural network through graphs using the Matplotlib python library.

The above figures show the graph of the actual price versus the price predicted by different neural networks on the real stock market data and the artificial data generated by us. While analysing the graph we can see that each of our neural networks can predict the trends of the market i.e., whether the stock price would increase or decrease. The models can also accurately anticipate the stock's closing price, with the GRU neural network's prediction having a mean absolute percentage error of just 1.53%. The GRU neural network was the top performer for the artificial data as well, with a mean absolute percentage error as low as 0.89%.

7. Conclusion

The main aim of our study was to solve the stock market prediction problem using neural networks and compare each neural network's performance. We were able to get the desired results from our methodology with the multi-layered GRU neural network being the most efficient of all. We utilised the knowledge gained during the literature survey to increase the efficiency of our neural networks. Most of

the existing works compare the performances of deep learning algorithms with traditional machine learning algorithms and have found that deep learning algorithms are more efficient than traditional machine learning methods. Therefore, our study compares different deep learning methods and suggests an efficient neural network for making stock market predictions.

A standout feature of our study is that we generate artificial stock market data using a generative adversarial network model. The data we generated is similar to real stock market data and has increasing and decreasing trends. In a time series forecasting problem, the data should be recent so that the model can make predictions based on recent trends and not on some long time ago and outdated trends. Therefore, we were limited with how much data we can use to train our neural networks. Having less amount of data can also lead to overfitting of the model as neural networks run many iterations. We were able to provide a solution for this problem as we generated artificial data and even tested our neural networks on this data.

Moreover, during the course of our study, we have conducted a literature review to gather constant feedback and to find potential gaps or opportunities in the current existing work which we can address with our proposed methodology and implementation. In the future, we can make predictions over the next week or even a month without compromising the efficiency of our models. As of now we just used the historical prices of Tata Steel only but in the future, we can use the historical prices of different companies and check how our model performs on a different dataset. Further, we can also use some more financial information like the PE or PB ratio, quarterly profits, and even news sentiments for predicting stock prices.

Declarations

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No funding was received for conducting this study.

Author Contributions

All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Raunak Shukla under the guidance of Poornima N. The first draft of the manuscript was written by Raunak Shukla and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Conflict of Interest

The authors have no competing interests to declare that are relevant to the content of this article.

Data Availability Statement

The data used in this research study is the historical share price of Tata Steel, which was obtained from Yahoo Finance. The data was accessed through the yFinance Python library, which provides a convenient interface to retrieve financial data for registered stocks using their unique identifiers. The yFinance library and Yahoo Finance platform offer public access to financial data for various stocks, including Tata Steel and the data can be accessed through the ticker of a company which is unique for each company. For Tata Steel, the ticker is 'TATASTEEL.NS'. We encourage interested researchers to refer to the yFinance documentation (<https://github.com/yahoo-finance/yahoo-finance>) for further guidance on data retrieval and usage.

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Figures

Model: "sequential"

Layer (type)	Output Shape	Param #
gru (GRU)	(None, 14, 50)	7950
dropout (Dropout)	(None, 14, 50)	0
gru_1 (GRU)	(None, 14, 50)	15300
dropout_1 (Dropout)	(None, 14, 50)	0
gru_2 (GRU)	(None, 14, 50)	15300
dropout_2 (Dropout)	(None, 14, 50)	0
gru_3 (GRU)	(None, 50)	15300
dropout_3 (Dropout)	(None, 50)	0
dense (Dense)	(None, 1)	51
Total params: 53,901		
Trainable params: 53,901		
Non-trainable params: 0		

Figure 1

Architecture of GRU Neural Network

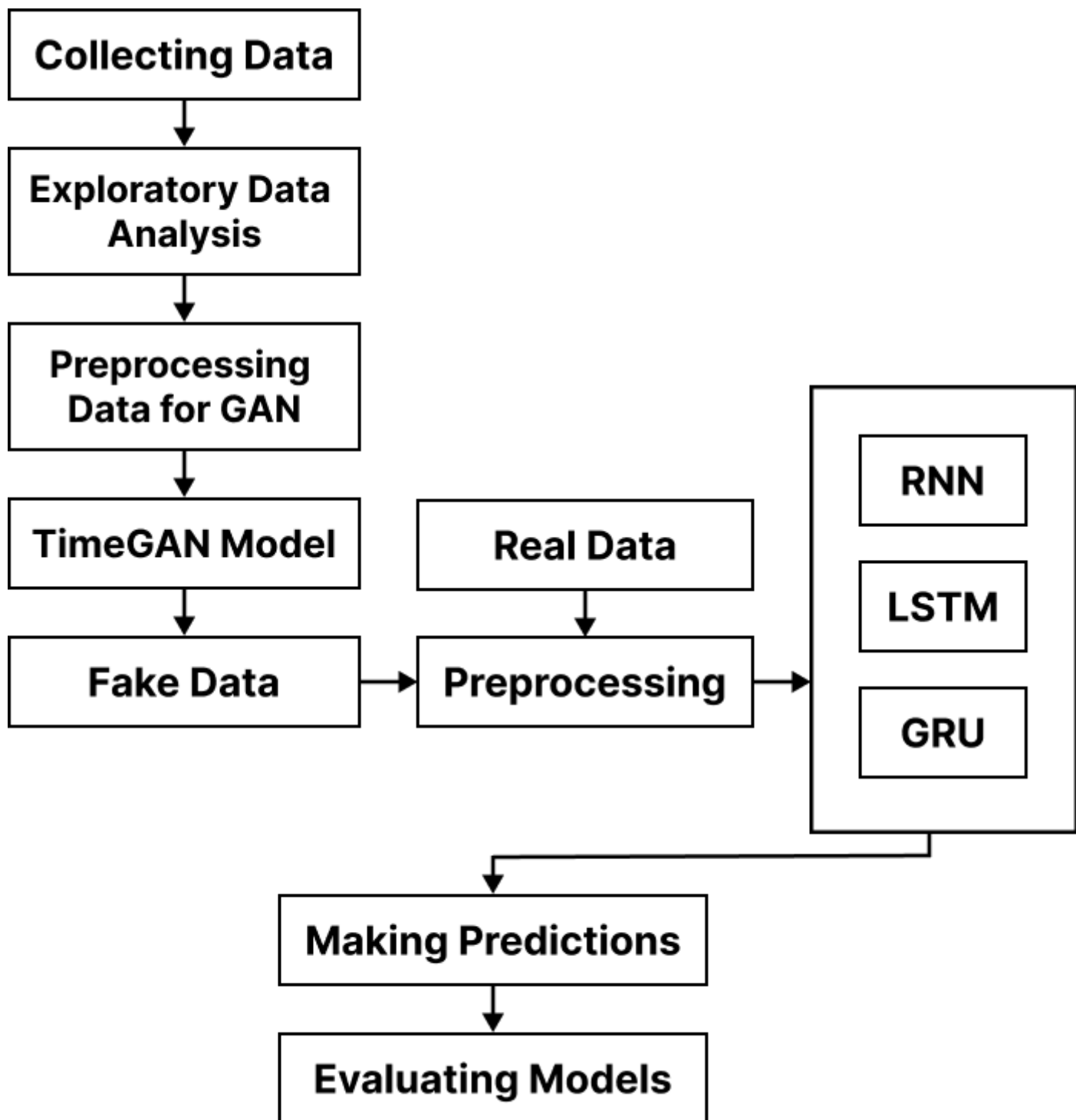


Figure 2

Architectural Block Diagram

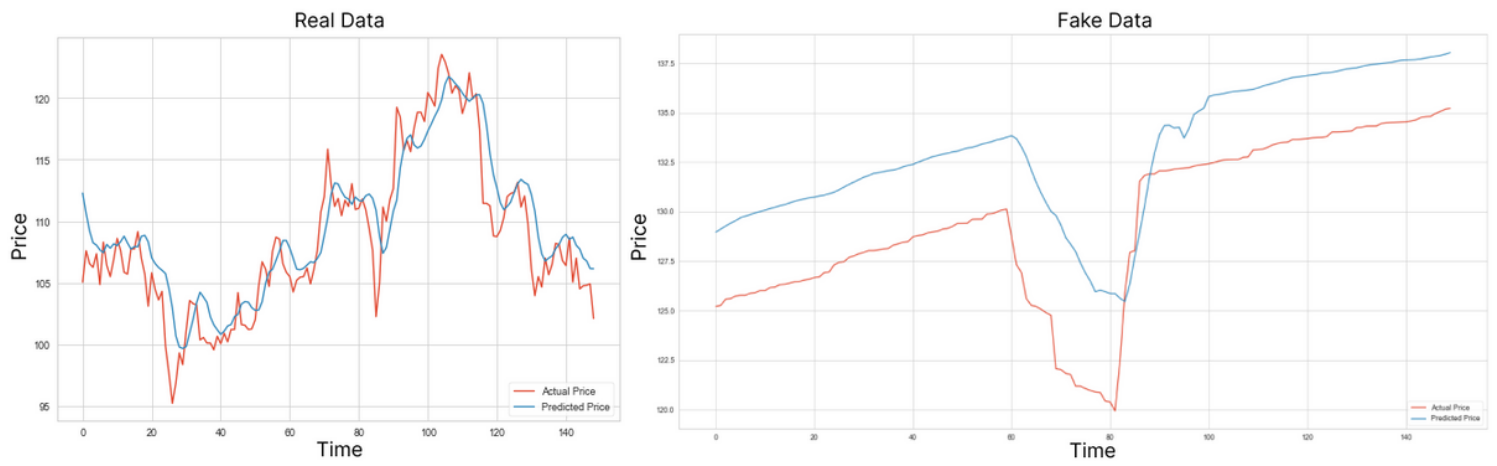


Figure 3

Actual Stock Price vs Stock Price Predicted by RNN

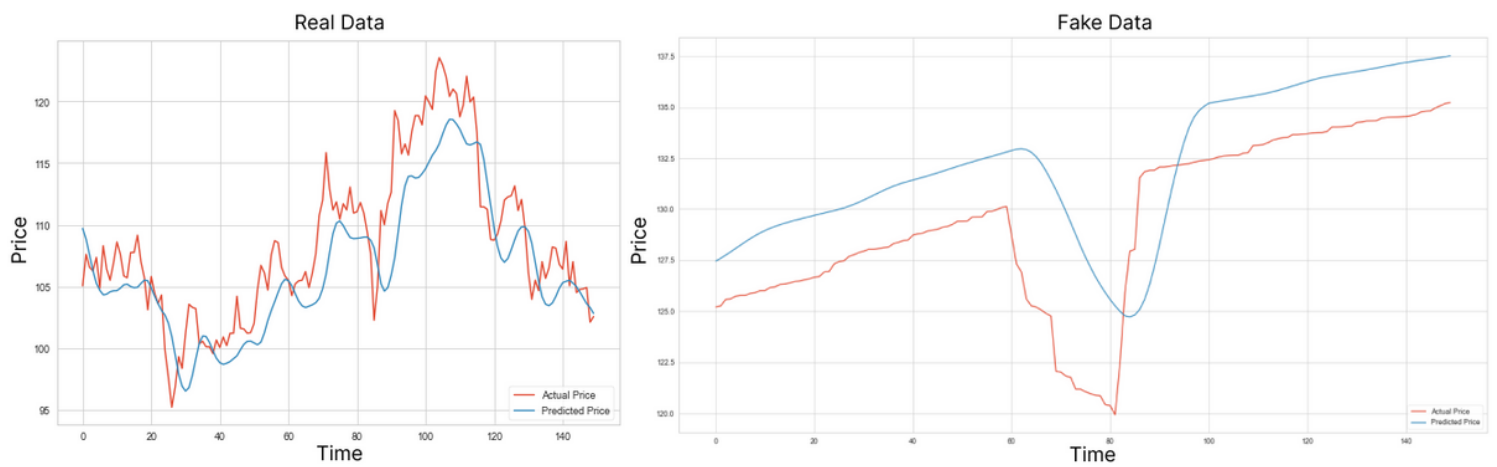


Figure 4

Actual Stock Price vs Stock Price Predicted by LSTM

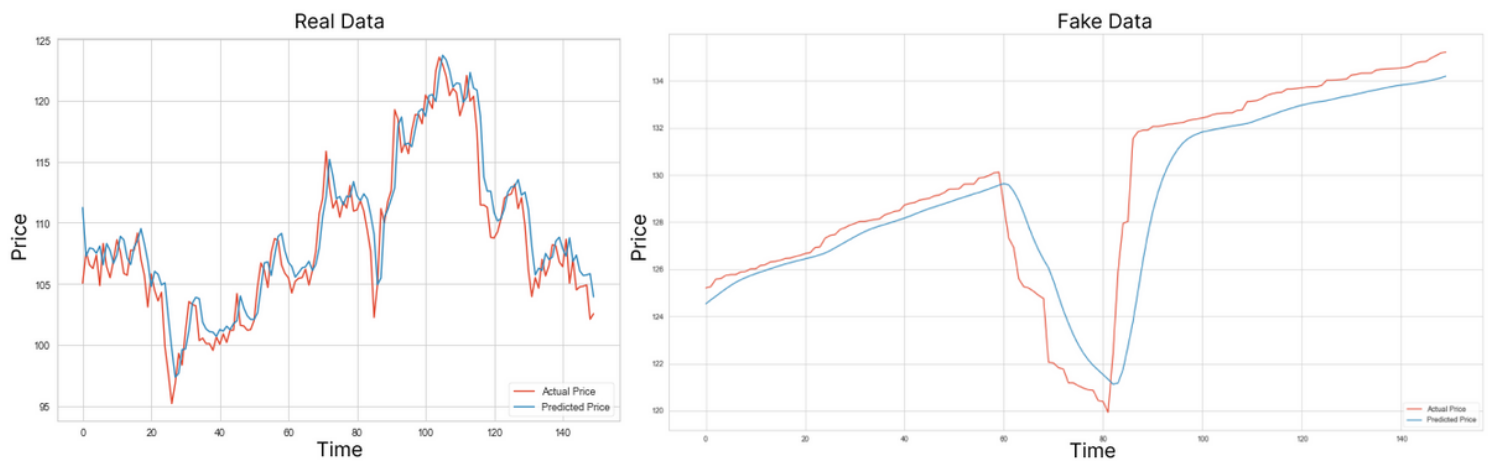


Figure 5

Actual Stock Price vs Stock Price Predicted by GRU

